



# Drafting Technical PRDs

Week 3 • Day 1 • Requirements Engineering & Mini-Capstone

TPM Academy



# **This Week is Non-AI**

The discipline of structured thinking expressed as  
written words

AI fills vagueness with confident generic prose. Vagueness is the bug  
we're eliminating.

# Day 1 Objectives

- Pick **one** feature concept from your Week-2 set, with explicit reasoning
- Write a **Context** section that gives an engineer a reason to care before reading what to build
- Write a **Problem** section grounded in your Week-1 interviews and Week-2 journey map
- State **goals + non-goals** as user outcomes, not feature names
- Write a **solution sketch** that lets an engineer imagine the shape without specifying the implementation



# Today's Shape

| Clock         | Block                       | Outcome                                         |
|---------------|-----------------------------|-------------------------------------------------|
| 09:00 – 09:15 | Opening                     | Pin Week-2 artifacts; restate the non-AI rule   |
| 09:15 – 10:30 | Block 1 + Activity 1        | Pick the one feature                            |
| 10:45 – 12:00 | Block 2 + Activity 2        | Section 1 (Context) + Section 2 (Problem)       |
| 13:00 – 14:15 | Block 3 + Activity 3        | Section 3 (Goals/non-goals) + Section 4 (Scope) |
| 14:30 – 16:00 | Block 4 + Activity 4 + Wrap | Section 5 (Solution sketch)                     |

# Block 1 — What a TPM PRD Is

And how it differs from a generic PM PRD

# TPM PRD vs Generic PM PRD

| Generic PM PRD                 | Technical PRD (your job)         |
|--------------------------------|----------------------------------|
| Heavy on business framing      | Engineer-readable from Section 1 |
| Vague on edge cases            | Granular AC + NFRs               |
| Light on dependencies          | Names systems, owners, gaps      |
| "Out of scope" usually missing | Out-of-scope explicit            |
| Risks listed as "low"          | Real risks named honestly        |

**The TPM contract:** a Product Requirements Document (PRD) an engineering lead would accept as scoping input *without a clarifying call*.



# The 11-Section PRD Anatomy

## This week's daily build

1. Context
2. Problem
3. Goals & non-goals
4. Scope (in / out)
5. Solution sketch
6. Acceptance Criteria (*Day 2*)

## Continued

7. Non-Functional Requirements (NFR) (*Day 3*)
8. Metrics & validation (*Day 4*)
9. Risks & open questions (*Day 4*)
10. Dependencies (*Day 4*)
11. Out-of-scope follow-ups (*Day 4*)

# Pick One Feature, Defend the Choice

Your Week-2 set has 3 feature concepts. Today you pick one. Use the four-dimension scoring sheet:

| Dimension         | 1                     | 5                                             |
|-------------------|-----------------------|-----------------------------------------------|
| Friction severity | Mild annoyance        | Task failure / lost data                      |
| Metric movability | Indirect, slow        | Directly moves a Tier Sheet metric in 30 days |
| Build feasibility | >1 quarter            | Achievable in 6–8 weeks                       |
| Learning value    | Confirms what we know | Tests an unproven assumption                  |

**Tie-breaker:** if two features tie, the quad must *argue*, not split the difference. The discipline of choosing is the work.



# Activity 1 — Pick the One Feature

Quads • 45 minutes

Score your 3 Week-2 feature concepts on 4 dimensions. Pick one. Write a "Why this one" memo: 2 paragraphs, user terms + business terms.

- 5 min: re-read the 3 cards aloud
- 20 min: score each on 4 dimensions
- 10 min: pick + argue if tied
- 10 min: write the memo



5 read • 10 score • 10 pick • 10 memo

# Block 2 — Context & Problem

Sections 1 and 2 of the PRD



# Section 1 — Context ( $\frac{1}{2}$ page max)

## What it answers

- Why this work?
- Why now?
- What changed?

## What it avoids

- "Customers are asking for it" (vague)
- Marketing language
- Restating company strategy from scratch

**The test:** after reading Section 1, the engineer should be willing to keep reading Section 2.



## Section 2 — Problem ( $\frac{1}{2}$ to $\frac{3}{4}$ page)

The **job-to-be-done**, today's **friction**, why **workarounds** fall short.

Tie explicitly to the **Week-2 journey map**. Reference your friction stars by name.

### ✗ Generic

*"Dispatchers struggle with reconciliation. The current process is inefficient and error-prone."*

### ✓ Specific

*"Dispatchers spend ~45 min after-shift cross-checking 3 systems. Maria (Interview A3): 'I just guess on the last 4 entries by 6:30 — I want to get home.'"*

# Activity 2 — Section 1 + Section 2

Quads • 50 minutes

Each member drafts Section 1 alone, then quad picks the strongest. Repeat for Section 2. Quote a real customer at least once.

- 20 min: solo Section 1 drafts
- 10 min: pick + combine
- 10 min: solo Section 2 drafts
- 10 min: pick + combine + quote check



20 + 10 + 10 + 10

# Block 3 — Goals, Non-Goals, Scope

Sections 3 and 4 — bound the work

## Section 3 — Goals (3–5)

- Each goal is a **user outcome**, not a feature
- Each goal is **observable within 30 days** of ship
- Each goal ties to a **Tier Sheet metric** from Week 2

### Feature-shaped

"Ship the new reconcile dashboard"

### Outcome-shaped

"Dispatchers see open work without hunting; median time-to-close drops 30%."

# **Non-Goals (2–4) — the under-valued section**

Non-goals state what you **explicitly will not do**. Done well, this section pre-empts most scope-creep negotiations later.

## **FieldPulse examples:**

- "We will not redesign the navigation bar"
- "We will not change the underlying data model for tickets"
- "This iteration will not support tablets"

**Coach's prompt:** "What's the most ambitious thing a stakeholder might think this includes?" Put that in non-goals.





## Section 4 — Scope (in / out)

| In scope                | Out of scope (this iteration) |
|-------------------------|-------------------------------|
| Mobile flow             | Tablet layout                 |
| Online + offline draft  | Offline-first sync            |
| English only            | Spanish localization          |
| Single-shop dispatchers | Multi-shop / franchise admins |

**The right column is the negotiation tool:** "I see what you might also want; here's why this iteration doesn't include it."

# Activity 3 — Section 3 + Section 4

Quads • 50 minutes

Draft 5 candidate goals; cull to 3. Draft 3 non-goals; cull to 2. Build the in/out scope table.

- 10 min: goals (draft 5, cull to 3)
- 10 min: non-goals (draft 3, cull to 2)
- 25 min: scope table
- 5 min: sanity-check goal ↔ metric ties



10 + 10 + 25 + 5

# Block 4 — Solution Sketch

Just enough for an engineer to imagine the shape



# What Goes In Section 5

## ✓ Include

- User-visible flow (4–8 steps)
- Key surfaces affected
- Hard interactions with other systems
- Happy-path narrative (one paragraph)
- Optional: 1–2 paper sketches

## ✗ Exclude

- Database schema decisions
- Specific application programming interface (API) contracts
- Class names, library picks
- Pixel-level layout

# ? The "Engineer's First Three Questions" Test

After writing Section 5, ask: what are the **first three questions** an engineer would ask after reading this?

| Their first question is about | Diagnosis                                 |
|-------------------------------|-------------------------------------------|
| Implementation choice         | You over-specified; pull back             |
| User behavior or scope        | You under-specified; add detail           |
| Edge cases / error states     | That's correct — AC tomorrow will address |

# Activity 4 — Section 5 Solution Sketch

Quads • 55 minutes + Wrap

Sketch the flow on paper. Write the happy-path paragraph. List hard interactions. Run the three-questions test.

- 25 min: sketch flow (4–8 steps)
- 10 min: happy-path paragraph
- 10 min: hard interactions list
- 10 min: three-questions test



25 + 10 + 10 + 10 • 15-min wrap-share



# Day 1 Wrap

## What we built

- Feature locked in with reasoning
- Section 1 Context
- Section 2 Problem
- Section 3 Goals + non-goals
- Section 4 Scope
- Section 5 Solution sketch

## What opens tomorrow

- **Granular Acceptance Criteria**
- Given/When/Then form
- The 5 acceptance criteria (AC) failure modes
- Peer-AC review

**Bring to Day 2:** Your draft Sections 1–5. AC will be written against them.

# Questions & Reflection

If an engineer read your Section 5 right now, what would their first question be?